



Olanipekun Adekunle

Senior Backend /
Infrastructure Engineer

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SUMMARY

Senior Backend Engineer with 6+ years building and scaling fintech and banking systems. Deep expertise in Node.js, TypeScript, distributed systems, and cloud-native infrastructure. Led payment processing platforms, core banking systems, and identity verification products processing 10K+ daily transactions across thousands of users. Proven track record owning backend architecture end-to-end and delivering at scale.

SKILLS

Languages & Frameworks

- TypeScript, JavaScript, Python, Golang, Node.js
- Microservices, Distributed Systems, REST APIs, gRPC, WebSockets,
- API Gateway Design, Clean Architecture

Databases & Storage

- PostgreSQL, MongoDB, MySQL, MSSQL, Database Migration & Optimization

DevOps & Infrastructure

- Kubernetes (Production), Docker, CI/CD (GitLab/GitHub Actions),
- ArgoCD (GitOps), Infrastructure as Code, Self-Managed Cloud,
- Monitoring (Prometheus, Grafana)

Cloud Platforms

- AWS, Azure, On-Premise Infrastructure (Linux and Windows)

Fintech & Domain Expertise

- Payment Systems, Core Banking, KYC/Identity Verification,
- Transaction Processing, Financial Compliance (CBN),
- Bill Payment Integration
- Real-Time & Performance
- WebRTC, WebSockets, High-Throughput Systems,
- Concurrency Patterns, Performance Optimization

Leadership & Practices

- Technical Leadership, Code Review & Mentoring,

ACADEMIC BACKGROUND

BACHELOR OF ENGINEERING

FEDERAL UNIVERSITY OF AGRICULTURE ABEOKUTA | 2023
NATIONAL SOCIETY OF ENGINEERS (NSE MEMBER).
NATIONAL YOUTH SERVICE CORPS (NYSC).

EXPERIENCE

STERLING BANK (HQ) Senior Backend Engineer

Marina Lagos, Nigeria
2025- present

- Architected Node.js KYC, identity verification, and payment processing systems integrating BVN, NIN, Interswitch, and Providus with OAuth 2.0, intelligent failover, and retry mechanisms — achieving 99.9% uptime
- Designed and executed auditable customer data migration pipelines for bank acquisition — onboarding 5,000+ customers with zero data loss, idempotent transaction handling, and full PostgreSQL optimisation
- Reduced transaction failure rate from 5% to 0.3% by owning incident response and root-cause analysis on critical payment systems implemented permanent fixes using Promise.all, worker threads, and event-driven concurrency patterns, improving overall reliability by 40%
- Built production-grade partner API integrations with rate limiting, webhook processing, circuit breakers, and automated failover maintaining system availability during third-party disruptions
- Architected RBAC system with Node.js and PostgreSQL for sensitive financial data, covering audit logging, admin action tracking, and CBN compliance controls
- Implemented multilayer transaction approval on the admin dashboard, preventing fraudulent actions and enforcing maker-checker controls across staff operations
- Built internal admin tooling and automated customer support workflows cutting manual operational intervention by 80%

NEPTUNE MICROFINANCE BANK. Senior Backend/ Lead Infrastructure Engineer

Ikeja Lagos, Nigeria
2023 - 2025

- Co-architected microservices Core Banking Application (CBA) across Authentication, Transactions, Loans, Investments, and Notifications processing 10K+ daily transactions on PostgreSQL, MongoDB, and Redis
- Designed production API Gateway exposing core banking via gRPC and REST with OAuth 2.0, rate limiting, and request validation enabling secure integrations for 15+ external partners
- Built E-Channels service for payment partner communication with mTLS, webhook processing, and async message queuing decoupling core banking from third-party failure cascades
- Delivered high-throughput Node.js notification service processing 3,000+ daily alerts across Email, SMS, Push, and WebSockets with dead letter queues and 99.8% delivery rate
- Engineered identity verification and bill-payment services integrating BVN, NIN, Interswitch, and Providus with circuit breaker patterns and intelligent provider failover
- Led full infrastructure build-out: self-managed Kubernetes clusters, Terraform IaC, GitLab/GitHub Actions CI/CD, HA PostgreSQL with HAProxy load balancing, and Wazuh SIEM saving \$50K+ annually
- Deployed high-availability observability stack (Prometheus, Grafana, Loki) with alert rules triggering on critical failures; self-managed Grafana and Prometheus with zero third-party dependency
- Executed zero-data-loss legacy CBA migration and delivered Web Banking platform serving 500+ corporate accounts with KYC workflows and RBAC
- Mentored junior backend and DevOps engineers through code reviews, architecture guidance, and hands-on infrastructure training

SLOCX. Senior Backend Engineer (Team Lead)

Arbedeen, United Kingdom
2021- 2023

- Led a team of 9 backend engineers delivering a multilingual learning platform across 5 countries, owning architecture decisions, code reviews, and engineering standards for security, performance, and scalability
- Designed and delivered RESTful APIs handling 50K+ daily requests using clean architecture principles maintained sub-100ms p95 response times under concurrent load
- Built real-time communication infrastructure using WebSockets for live chat and WebRTC for peer-to-peer video streaming, achieving sub-200ms latency across multiple regions
- Implemented CI/CD pipelines with GitLab CI and Docker, cutting release cycles from days to hours; introduced ELK stack logging and Prometheus/Grafana dashboards that reduced incident response time by 60%
- Architected backend concurrency model using goroutines and channels to handle high-traffic volumes and real-time connections without service degradation
- Mentored 6 engineers on backend best practices, establishing team-wide standards that improved code review turnaround and reduced production incidents

COLLAB CARDS.

 **United Kingdom**

Backend Engineer (BACKEND)

2020- 2021

- Architected Node.js backend for a social content platform (writers, poets, fiction) built content posting, likes, comments, and idea-sharing features on PostgreSQL with Docker and AWS
- Designed high-performance REST APIs with Express.js following clean architecture principles covered user management, bookings, and real-time chat services with consistent sub-100ms response times
- Built real-time messaging infrastructure using Socket.io for low-latency communication, handling concurrent connections efficiently via Node.js event-driven, non-blocking I/O and worker threads
- Optimised PostgreSQL query performance and Redis caching strategies reduced latency and improved throughput across multiple regions under high concurrent load
- Integrated third-party APIs for payment processing, analytics, and email notifications; implemented CI/CD pipelines with GitLab and Docker, automating testing and cloud deployments
- Introduced centralised observability with Winston/Pino logging, metrics collection, and alerting — improving system health visibility and reducing incident response time
- Built comprehensive automated test suites with Jest and Supertest; conducted cross-functional code reviews with frontend, product, and QA teams ensuring production-ready, secure deliverables
- Mentored junior engineers and established Node.js backend standards across security, performance, and scalability — drove team-wide adoption of best practices

KHAIME AI

 **Texas, USA**

Team Lead (Backennd)

2019 - 2020

- Inherited a poorly architected codebase — restructured data models, refactored database schemas, and introduced clean modular architecture optimised for scalability and long-term maintainability
- Designed and executed large-scale data transformation and migration pipelines with full backups, ensuring zero data loss and consistency across restructured schemas
- Built a recurring payment scheduling system enabling clients to automate payments for goods and services handled edge cases, failure retries, and idempotent execution
- Decomposed monolithic backend into independently deployable microservices enabling per-service scaling and faster release cycles
- Optimised high-traffic API endpoints, reducing response times by 60%+ through query tuning, caching strategies, and async processing improvements
- Participated in stakeholder planning sessions, translating business requirements into technical roadmaps and delivering against them as team lead
- Mentored junior developers on backend architecture, coding standards, and engineering workflow establishing team-wide best practices adopted across the project

PROJECTS

GOMONEY

<https://play.google.com/store/apps/details?id=com.global.gomoney>

Contributed to the development of Gomoney, a modern fintech app offering seamless bill payments, peer-to-peer transfers, virtual cards, budgeting tools, transaction categorization, spending analytics, and in-app customer support — designed to enhance user financial management and digital banking experiences.

NVAULT

<https://play.google.com/store/apps/details?id=com.neptune.nvault&hl=en>

Contributed to the development of nVAULT, a cutting-edge fintech app at Neptune Microfinance Bank Ltd., revolutionizing traditional banking with an all-in-one digital platform for payments, savings, and investments. Played a key role in designing a streamlined digital onboarding process, enhancing user experience, and optimizing savings and investment features to drive user engagement and financial empowerment.

QUIZMEAPP

<https://play.google.com/store/apps/details?id=com.quizme.quizme>

Led the development of a rewards-based gaming app where users earn by answering questions, gaining points, and competing on a leaderboard. Features included cash withdrawals, point redemption for airtime and data, and real-time performance tracking.

ZERO PAY

<https://play.google.com/store/apps/details?id=com.myzeropaypay.app>

Led the Development a real estate application with flexible installment payments, secure wallet integration, and automated direct debit mandates. The app also supports utility payments (airtime, data, electricity), offering users a seamless property purchase and financial management experience.

SLOCX

<https://slocx.com>

Lead a Team of 9 to develop online language center for learning languages in over five countries offers various features, such as live chatting, live video lessons with tutors, lesson booking, and more. The backend is built using Golang, a highly performant and scalable language ideal for handling concurrent tasks. The web framework Chi is used to develop lightweight and fast REST APIs for handling various functionalities, such as user management, lesson booking, and chat services.

For real-time communication, such as live video lessons and messaging, WebSockets are utilized on the backend. They facilitate a persistent connection between the server and client, ensuring low-latency communication essential for interactive features.

On the frontend, WebRTC is integrated for seamless, peer-to-peer video streaming, allowing students to engage in live video lessons with their tutors. This combination of technologies ensures a smooth, real-time interactive learning experience across multiple regions.

BEAST MOOD.

beastmood-anime.vercel.app

Led a Team of Backend Engineers in developing an **anime streaming platform** that offers features like real-time video streaming, user comments, and the ability to like content. The platform is built with **Node.js** and **Express** to handle server-side logic, while **MongoDB** provides flexible and scalable data storage for user interactions, such as comments and likes. The project is hosted on **AWS**, ensuring reliable performance and scalability. Although the core functionality is in place, the platform is still in development, with additional features and optimizations being worked on to enhance the user experience.